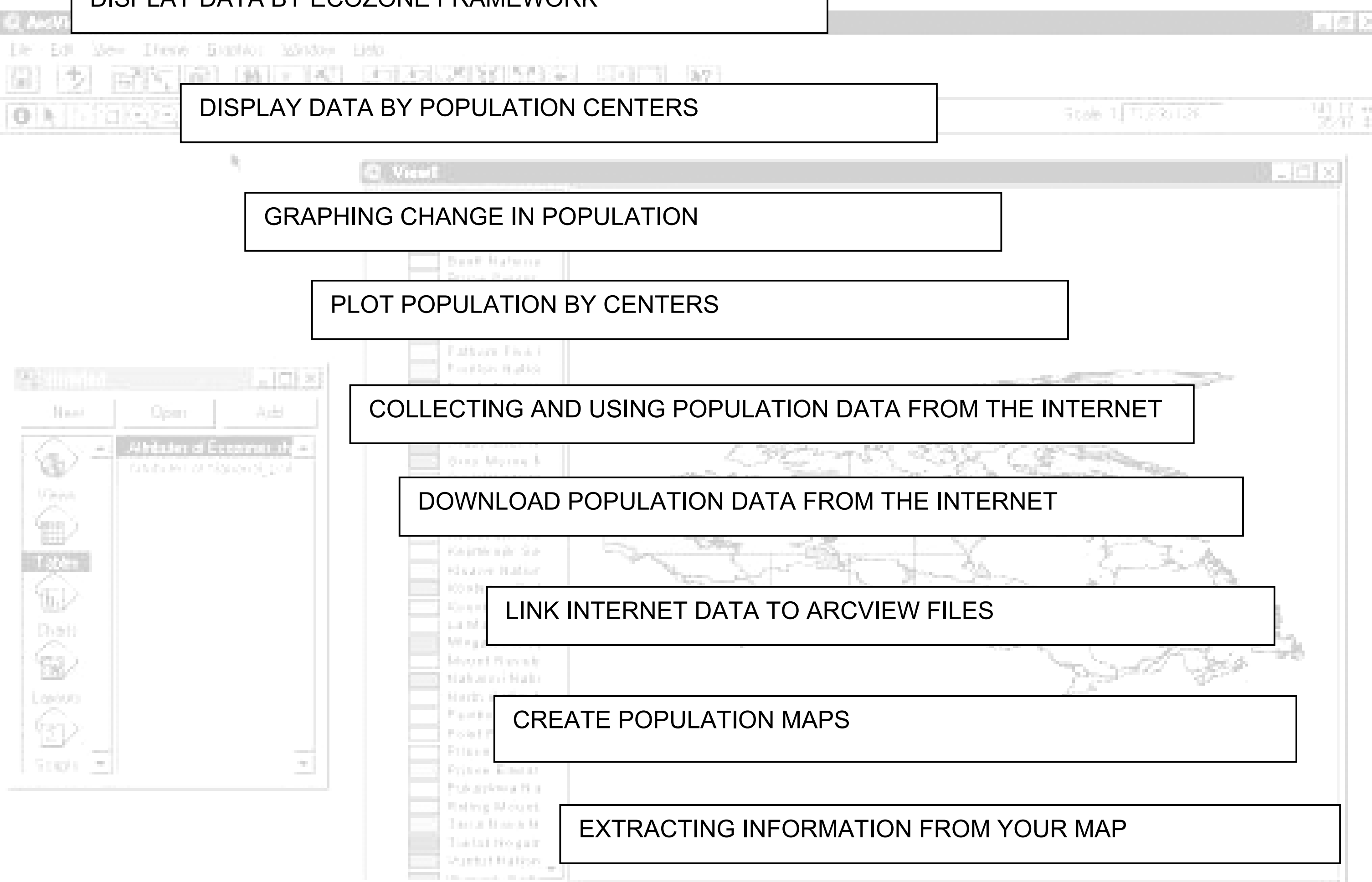
Canada's Population Trends



DISPLAY DATA BY ECOZONE FRAMEWORK

DISPLAY DATA BY POPULATION CENTERS

GRAPHING CHANGE IN POPULATION

PLOT POPULATION BY CENTERS

COLLECTING AND USING POPULATION DATA FROM THE INTERNET

DOWNLOAD POPULATION DATA FROM THE INTERNET

LINK INTERNET DATA TO ARCVIEW FILES

CREATE POPULATION MAPS

EXTRACTING INFORMATION FROM YOUR MAP

GEOMATICS FOR HIGH SCHOOL CURRICULUM

ArcView Lesson

GIS Information

Here's a list and explanation of important features in ArcView:

Project : In ArcView, a file for organizing your work. Projects use five types of documents to organize information: views, tables, charts, layouts, and scripts.

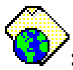
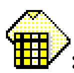
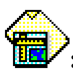
View  **:** A component of an ArcView project used for displaying, querying, and analyzing geographic themes.

Table of Contents: All themes in a View are listed to the left of the map. The Table of Contents shows the symbols used to draw features in each theme. The check box next to each theme indicates whether or not it is turned on or off in the map (whether it is currently drawn on the map).

Theme: A set of related geographic features, such as streets, parcels, or rivers, and the characteristics (attributes) of those features.

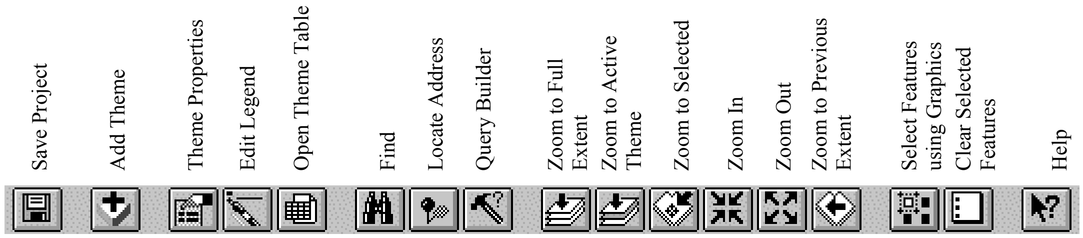
Tables  **:** Information formatted in rows and columns. A component of an ArcView project used for displaying Tabular data.

Layout  **:** The design or arrangement of elements in a digital map display or printed map. A component of an ArcView project used for creating presentation-quality maps.

Hotlink: In ArcView, a way to display data (for example, a file, image, ArcView document, or project) directly from a view, by clicking on a feature.

ArcView Buttons Reference Guide

View - Buttons



View – Tools

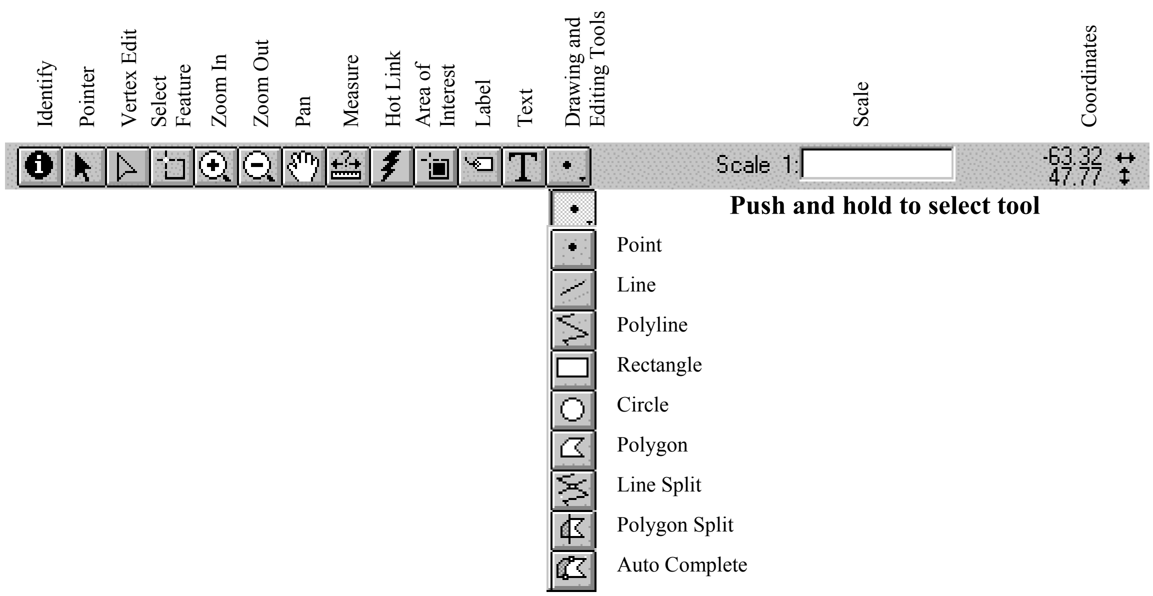


Table - Buttons

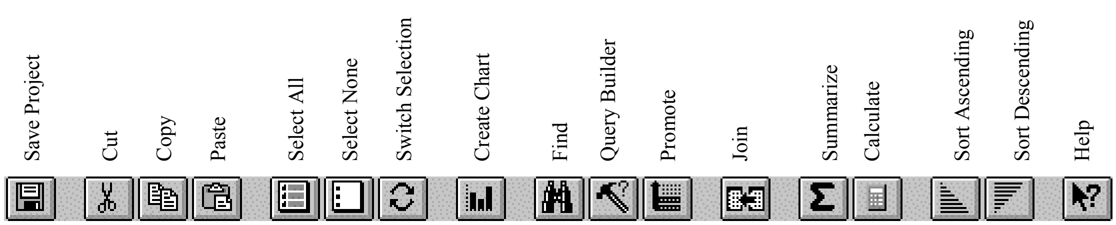


Table - Tools

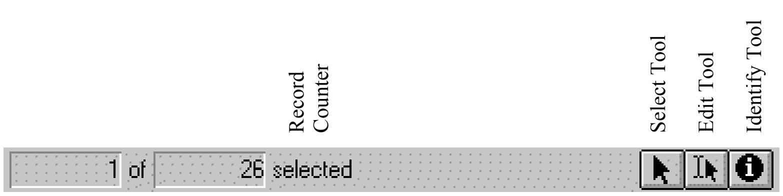


Chart - Buttons

Save Project
Undo Erase
Area Chart Gallery
Bar Chart Gallery
Column Chart Gallery
Line Chart Gallery
Pie Chart Gallery
Scatter Chart Gallery
Chart Properties
Series From Records/Fields
Find
Help

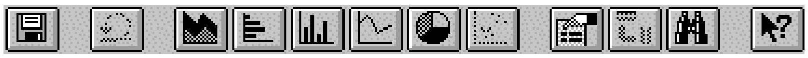
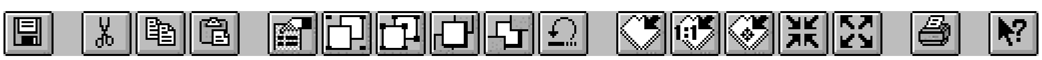


Chart - Tools

Identify Tool
Erase Tool
Erase Within Polygon Tool
Chart Element Properties Tool
Chart Colour Tool

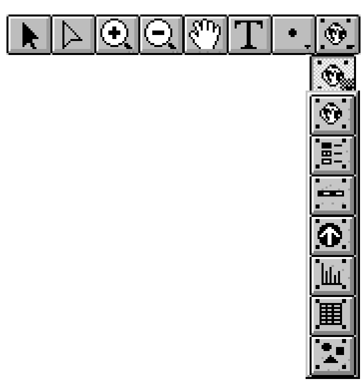
Layout - Buttons

Save Project
Cut
Copy
Paste
Layout Properties
Group
Ungroup
Bring to Front
Send to Back
Undo
Zoom to Page
Zoom to Actual Size
Zoom to Selected
Zoom In
Zoom Out
Print
Help



Layout - Tools

Pointer Tool
Vertex Tool
Zoom In Tool
Zoom Out Tool
Pan Tool
Text Tool
Drawing Tools
Frame Tools



Push and hold to select tool

View Frame
Legend Frame
Scale Frame
North Arrow Frame
Chart Frame
Table Frame
Picture Frame

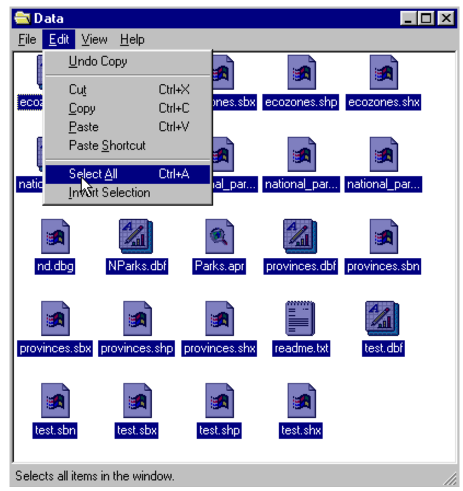
For more Information: http://www.esricanada.com

Getting Started

Before you begin, the lesson data folder must be ***copied*** into a drive in which the student has writing and editing capabilities. This will probably be the drive containing your own personal account on the school's system or it may be the **C: drive** of the machine you are currently working on. If the latter is the case you will have to work on the same machine each day until this assignment is complete. Each school set-up is different so work with your teacher to prepare your data folder.

! *(The data for the lesson must first be downloaded from this Web site and placed on the school computer or network. Ask your teacher where to find the data folder that has been downloaded).*

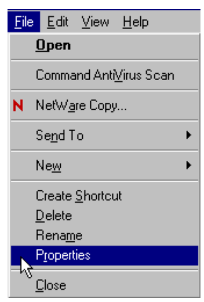
To do this, use the file manager program on your computer. For windows based systems this will be **My Computer / Windows Explorer /.**



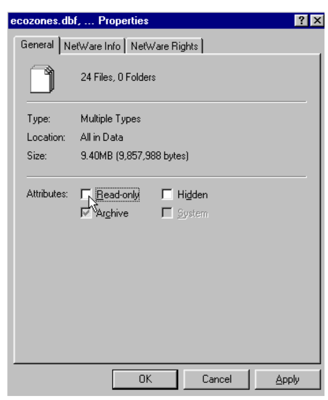
Once the data folder has been ***copied*** to the required destination the files must be made usable by removing the ***read only*** attribute. For windows based systems follow these instructions:

! *(The data downloaded from the Web Site is set as **read only** so that it cannot be changed or altered by accident.)*

1. Open your new data folder
2. Choose **Edit >** ***Select all*** (they will all appear Blue)
3. From the menu choose:
   File > Properties



4. Remove the check mark in the box beside the label ***read only***.



5. Click **apply**, then **OK**

The data in your new data folder can now be opened, edited and saved using ArcView or other GIS software that can use SHAPE files to carry out the following project.

! Denotes an observation.

? Denotes a question to be answered on the Student Activity Sheet.

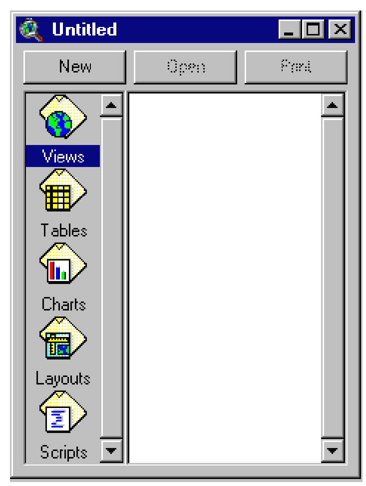
STARTING THE PROJECT

The purpose of this lesson is to use population data in various ways to show how it relates to different frameworks. A framework is the structure that is used to group data for information purposes. Examples of frameworks are by province, by ecozone, by city. In this lesson we will display attributes of population by using variable symbols to denote size classification.

STARTING ArcView

Begin by starting up the **ArcView** program. If you are not sure how to do this, ask your teacher.

1. Choose *as a blank project*
2. Click **O.K.**



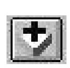
When the program starts, it automatically opens a project window titled **"Untitled."**

If you like, you can maximize the **ArcView** window to fill the screen.

! *In ArcView, you work within a project. A project is a collection of documents (map views, tables, charts, layouts, etc), document user interfaces, (called DocGUIs) and scripts. Only one project is open at a time.*

3. Now you are ready to add documents to your new project.

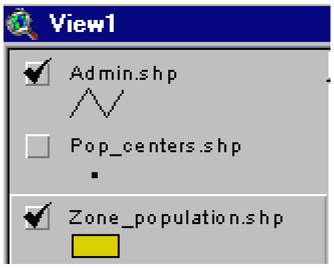
4. With the **"Views"** icon selected, click on **New** and maximize the **View1** window.

5. Click the **Add theme button**  and Navigate to the data folder where your data is stored. Hold down your **SHIFT** key and select the following files:

Admin.shp
Pop_centers.shp
Zone_population.shp

6. Click **O.K**.

This will add these three themes to the table of contents of the new view. They are now available for display and editing.



7. Click and drag to move the themes up or down in the table of content to change the order of display if necessary.

! Note the themes are displayed from the bottom to the top so line and point themes should be on top of polygon files to be visible.

8. **Click** on the square next to the themes you want to display. The theme that appears raised in the table of contents is the current active theme. This can be change by **Clicking** on other themes.

9. Remember to **Save** your project as you go along. Go to the **File** menu and select **Save Project.** Save your work to the appropriate directory (ask your teacher if you are not sure). Give your project a name that will identify your work. (The suffix **apr.** means that you are saving your work as a project that can be opened up in the future)

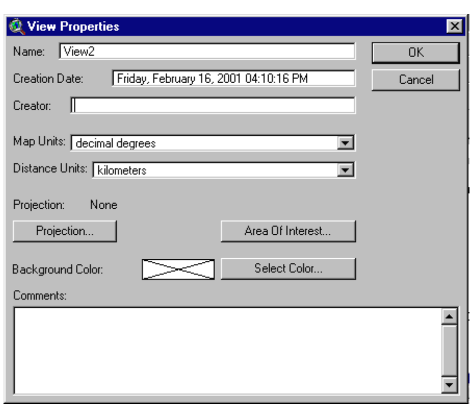
ArcView - Changing the Map Projection

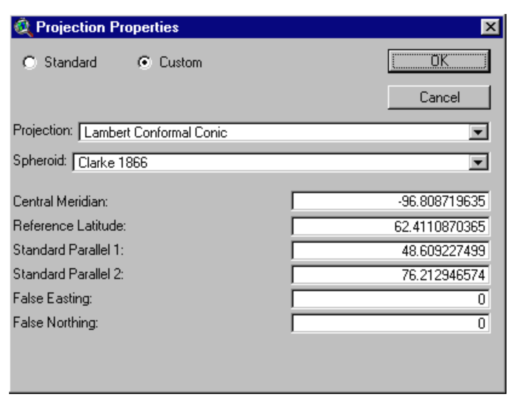
! A ***Map Projection*** defines the relationship between the map coordinates and the geographic coordinates, (latitude and longitude). Our current display shows what a map of Canada looks like with only a geographic projection.

CHANGING PROJECTION

Now we will give our map "projection coordinates" to make it look more like the maps which are familiar to us.

1. From the **View** menu, choose **Properties**.
2. Select **Properties.**
3. Select **Map Units** list. Change the **map units to** *decimal degrees.*
4. Select **Distance Units** list. Change the **distance units to** *kilometers*
5. Click the **Projection** button.
6. Select **Custom.**
7. Select **Lambert Conformal Conic** from the projection list.



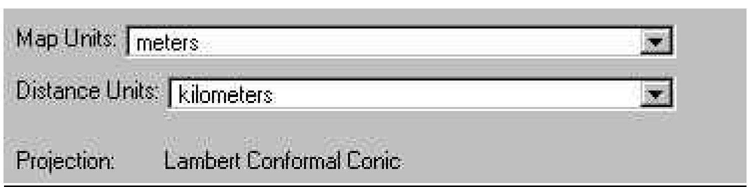


! Take a look at the parameters for this projection. Parameters are the descriptions of how the projection was made.
Note that a spheroid was used. (The earth is a slightly flattened sphere toward the poles).
Conformal implies that all angles are indicated correctly. All angles measured on the earth's surface are given the same values on the map.
Conical means in the shape of a cone. In this projection, the relative positions on a globe are transferred to a plane in the shape of a cone around the globe. See how the longitudinal lines are closer together at the top of the image, that is, they are not parallel.

The ***central meridian***, ***false easting*** and ***false northing*** are used to define the X (easting) and Y (northing) coordinates of the map. The ***reference latitude*** and ***standard parallel 1 and 2*** are used to determine the origin of the projection (i.e. the point from which you are viewing the globe).

8. Click **O.K.**

The bottom of the **View Properties** window should look like this:

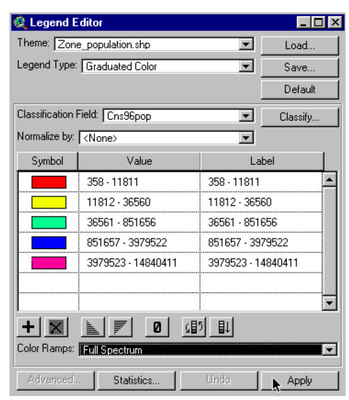


9. Click **O.K.**

! Note the difference in the map presentation.

? How is it different from the non-referenced map?

ArcView - DISPLAY POPULATION DATA BY ECOZONE FRAMEWORK



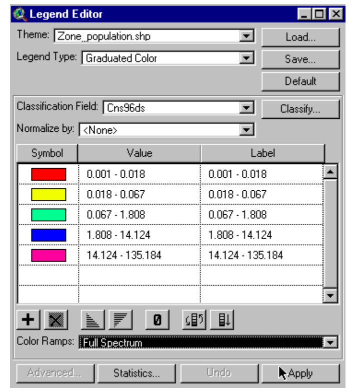
LETS LOOK 1996 POPULATION DATA.

1. Double Click ***Zone_Population*** theme.
2. Open **Legend Type** list, select **Graduated Colour**.
3. Open **Classification** field, Select ***Cns96pop***.
4. Open **Colour Ramps** list, Select **Full Spectrum**.
5. Click **Apply**.
6. Close **Legend Editor** window.

? What is the general pattern of population by ecozones?

? Does the area of the ecozone have an effect on its population?

NOW LET'S DISPLAY POPULATION DENSITY



Population density is the number of people per square kilometer.

1. Double Click ***Zone_Population*** theme.
2. **Classification Field** list, Select ***Cns96ds***.
3. Open **Colour Ramp** list, Select **full spectrum.**
4. Click **Apply.**
5. Close **Legend Editor** window.

? What is the general pattern of population density by ecozone?

? How is it different than 1996 population done above?

? Why?

NOW LET'S DISPLAY POPULATION CHANGE

Populations change and move over the years. Let's look at the change between 1971 and 1991.

1. Double Click ***Zone_Population*** theme.
2. Open **Classification Field** list, Select ***Pc71-91***
3. Open **colour ramp** list, Select **Full Spectrum**
4. Click **Apply.**
5. Close **Legend Editor** window.

? Why is it important to map changes in population?

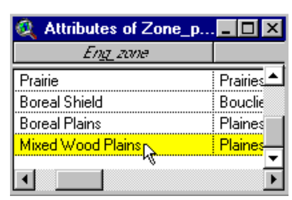
? Which ecozone has had the greatest change in population? Consider both increases and decreases.

? What implications does this have for the social, economic and environmental structure of the ecozone?

? Which ecozones have the next greatest population change?

ArcView - GRAPHING CHANGE IN POPULATION

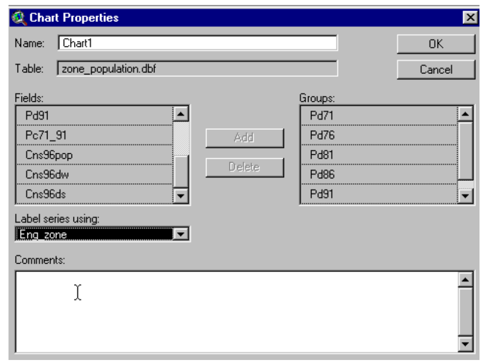
1. Click **Open Theme Table**  for the ***zone_population*** theme.



2. Click ***mixedwood plains*** ecozone record to select it.

3. Click **Create Chart** button

4. From **Fields** list, **select *Pd71, Pd76, Pd81, Pd86, Pd91, Cns96ds*** and Add To **Group List**.
5. Open **Label Series Using** list
6. Select ***Eng_zone***.
7. Click **O.K.**



? What does the chart tell you about the population growth in the Mixedwood Plains ecozone between 1971 and 1996?

8. Close **Chart 1** window.

9. Click **Select None** button. to clear all selected records.
10. Now Select **The *Pacific Maritime*** and ***Atlantic Maritime*** ecozones. (Hold shift key while selecting)

11.Click **Create Chart** button.

12. From the **Fields** list, **select *Pd71, Pd76, Pd81, Pd86, Pd91, Cns96ds*** and Add To **Group List**.
13. Open **Label Series Using** list, Select ***Eng_zone***.
14. Click **O.K.**

? What can you say about the population change in the two maritime ecozones?

15. Close **Chart 2** window.

? Select another ecozone or group of ecozones.
Write down what you expect to see in the charts for change in population density.
Produce the chart and compare it with what you expected.

? Are there differences in what you expected to see and the graph for the selected ecozone(s)?

? Why?

16. Close **The Table** window.

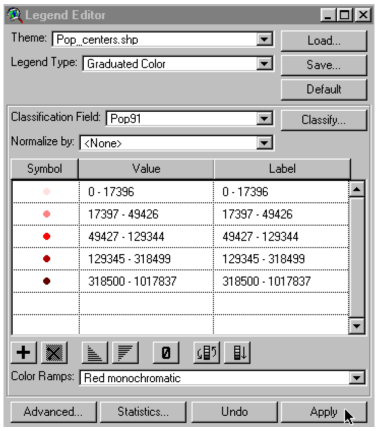
ArcView - PLOT POPULATION BY CENTERS OF POPULATION

1. Double Click ***Zone_Population*** theme.
2. Open **Legend Type** list.
3. Select **Single Symbol**.
4. Click **Apply**.
5. Close **Legend Editor** window.

6. Check ***Pop_centers*** theme to make it visible.

? What can you say about the distribution of population centres?
You may have to edit legend colour to optimize the visibility of the points.
You can zoom in to get a better look.

1. Double click ***Pop_centers*** theme.
2. Open **legend type** list.
3. Select **graduated symbols.**
4. Open **classification field** list.
5. Select ***pop91***.
6. Click **Apply.**
7. Close **Legend Editor** window.



? What can you say about the distribution of population centres based on population size?

Zoom in to get a good look.

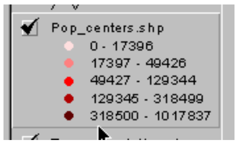
? How does this compare with what we discovered using ecozone data?

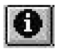
? Give pros and cons for using the two methods.

ArcView - EXTRACTING INFORMATION FROM YOUR MAP

Now that you have added data to your map, let's find the names of some of the population centers you mapped.

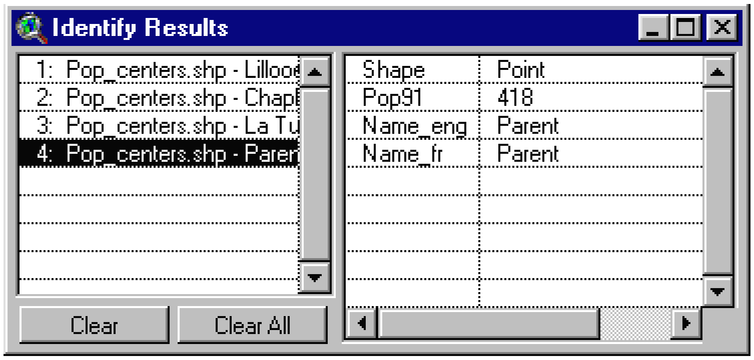
1. Click ***Pop_centers*** theme to make it active (it should appear raised).



2. Click **The Identify** button. 

? Using your cursor, find a large population centre near where you live

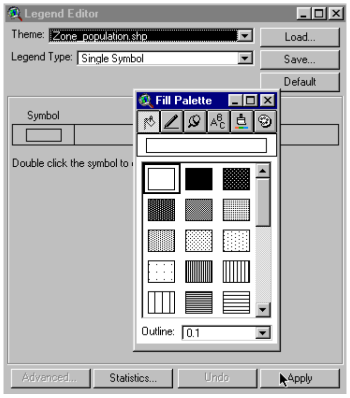
3. **Click**. This will give you an Identify Results table, which will give the name of the population center and the 1991 population.



? Using the cursor, find four other large population centres in Canada that are in ecozones different from your own.
Record the name of each city and the ecozone in which it is found.
Zoom in to locate population centers more easily.

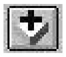
? How does the 1991 population data for the five centres compare? Can you draw a graph with this data?

COLLECTING AND USING POPULATION DATA FROM THE INTERNET

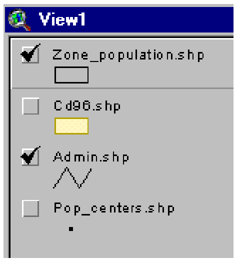


1. Double click ***zone_population*** theme.
2. Open **Legend type** list.
3. Select **Single symbol**.
4. Open the Fill Palette by double clicking symbol.
5. Select the clear palette to view only the outline.
6. Click **Apply**.
7. Close **Legend Editor** window.

8. Un-Check **Admin.Shp**. to turn it off .

9. Click the **Add theme button**  and Navigate to the data folder where your data is stored.
10. Select **Cd96.shp** file.
11**.** Drag And Drop the **Zone.Population.Shp** to the top layer of the view.

12. The layers of your view should now look like this.



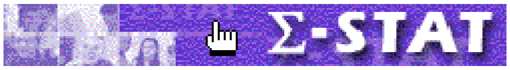
Accessing E-STAT on-line

1. **Access** the Internet version of E-STAT via Statistics Canada's web site at www.statcan.ca

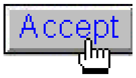
2. **Click** on the language of choice - English or French.

3. **Select Education Resources**.

4. **Choose** the link called **E-STAT**.

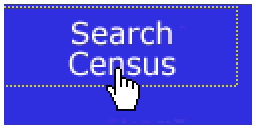


5. There are several resources available here. To access the data, select Accept (at the bottom of the page). You may need a password to connect.



This part of the exercise will explain how to obtain the census data from E-STAT Online and how to set it up so that you can bring it into ArcView later on. This portion of the exercise also assumes that you are using the Internet version of E-STAT 1999.

! Note: We are only going to use the Census Data portion of E-STAT since this is the only data that can be joined to existing data in ArcView.

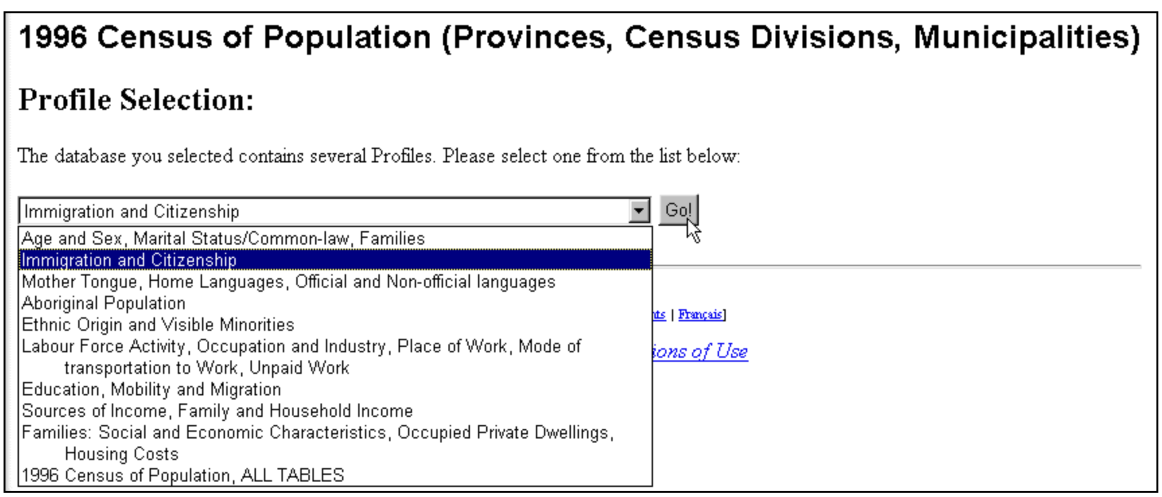
6. **Select** 

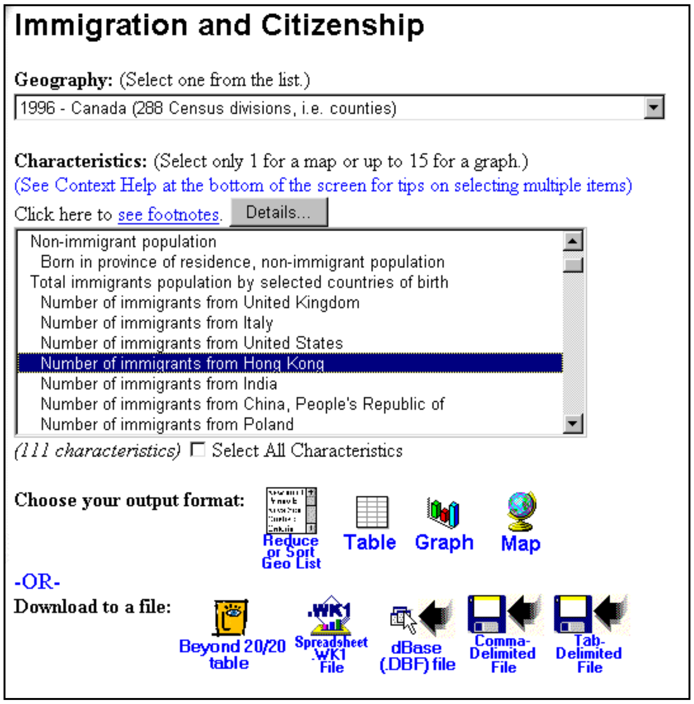
from the left bar menu.

7. At the **Available Databases** screen, Select The 1996 Census and then click on the **Go!** Button. (For this exercise we are going to use 1996 census information. However, the same concepts apply to the 1986 and 1991 censuses).

8. At the **1996 Census** screen, from the drop down list, select **1996 Census Of Population** (Provinces, Census Divisions, Municipalities) and click on the **Go!** Button. (Note: Municipalities are also called Census Subdivisions, and counties or regional districts are also called Census Divisions).

9. At the **Profile Selection** screen, from the drop down list select The Table **Immigration And Citizenship** then click on the **Go!** Button. (At this point you can select any profile from the list below that you are interested in, we are just using this profile as an example)





10. You are now at the **Geography and Characteristics** screen. From the **Geography**, click on the grouping of areas for which you want to extract data; e.g. **Census subdivisions in Ontario - 1996 - Ottawa (92 areas**). We are using **(1996 – Canada (288 Census division, i.e. counties)**

11. From the **Characteristics** drop down list, **select** the characteristics you would like to examine. **Scroll** through the list and decide what you want to look at, and then select your choices. **To select** a characteristic, **Simply Click On The Characteristic** of your choice, it will be highlighted. Continue to select the characteristics that you would like to examine. (Note: For a Windows-based browser, you can select multiple non-adjacent characteristics by holding the 'Ctrl' key while clicking on your selection.) (For a MAC-based browser, you would hold the 'Command" key while clicking on your selections.)

12. The next step is to **Choose Your Output Format**. For use in **ArcView GIS**, you will need to **Select The Table** icon to output the selected data in Table format. Your computer will think for a while and then the table will appear in your browser window.

13. Before you continue, it is very important that you: **Make Note Of The Characteristics** (column headings) and Countries (rows headings) that you chose and the order that they are listed in the table. It would be helpful to get a record of this information, especially the characteristics you have selected by printing the table. When you save your table, it will not save the field headings so you need to either print the table or write each one of them down so that you

can change them later. Furthermore, the names of the municipalities/counties that appear in the row (record) headings will not appear in the table either, only their unique code.

14. Once you have written everything down or printed the page, you are now ready to export your table. At the bottom of the page you will notice that there are several options. Next to the heading **"Download to a file:"** click on the **dBase (DBF)** file icon. You will be asked to SAVE the file somewhere and give it a name (make sure you remember where you saved it!).

15. When you are done exporting your table(s), you can close your E-STAT application. You are now ready to bring your data into ArcView and map it!

Using E-STAT data in ArcView.

We will now add the E-STAT data to our ArcView project.

1. Click and Check the **Cd96.shp** theme. You will make a map of Ontario's census subdivisions for 1996 appear.

2. **Click Once** on **Cd96.shp** theme to active it (it will appear raised).

3. Click on the **Open Theme Table**  button. You will see the attribute table for the census subdivisions appear.

4. From the **Window Menu, Choose Population.Apr** to go back to your Project Window.
5. Click on the **Tables Icon** and then click on the **Add** button. You are now going to add the table that you created in E-STAT.

6. In the **Add Table Dialog Box** navigate to the directory where you saved your E-STAT table.
7. **Select** the E-Stat table and then click on **OK**. That table will now appear in ArcView. Rearrange your windows so that they do not overlap.

8. Notice that in your table now, the field headings are not displayed, all that is listed is **"Field_1", "Field_2" etc**. You now need to set up what is called an **"alias"** for your table so that the proper field headings can be listed. Make sure that your E-STAT data table is the active one (if you are not sure, click on the title bar at the top of the window).
9. From the **Table Menu, Choose Properties**. In the dialog box that appears you will see the field headings listed. Under **"Alias"**, type in the fields' headings for your E-STAT table (Use the data you recorded when you were at the E-STAT web site).
10. When you are done, click on **OK.**

Visible	Field	Alias
✔	Geocode	
✔	Field_9	Italy
✔	Field_11	Germany
✔	Field_16	United States
✔	Field_18	England
✔	Field_56	Somalia

11. You will notice that your field headings are as they should be. You now need to join your E-STAT table to the table showing the census subdivision table so that your map has the data from E-STAT. In your E-STAT table, click on the field heading called Geocode (it will appear dark gray and pushed in).

12. From the **Window Menu,** choose **Attributes Of Csd96.Shp**. Scroll along until you find the field called E-STAT and click on the field heading (it will also appear pushed in).

13. Click on the **Join**  button.

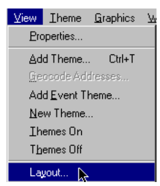
! Your E-STAT table will disappear and attach itself to the end of the Attributes of **Csd96.Shp** table. You are now ready to map your new data!

14. From the **Window Menu,** choose **View1**.
15. Double click on the **Csd96.Shp** theme to bring up the **Legend Editor**. You can now choose which kind of thematic map you would like to make and what classification field you would like to use!

ArcView - Map Layout for printing

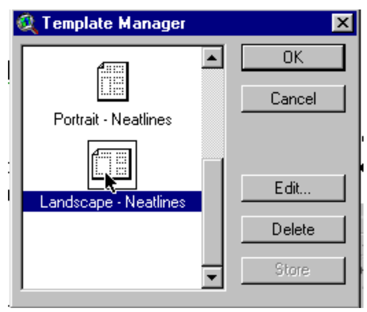
Once you have a map classified and displayed to your satisfaction, you should prepare a map layout that can be used for printing. The purpose of this lesson is to set up your map on a page and annotate it for presentation.

SET UP MAP



1. Click Menu **view.**
2. Click **Layout.**

3. Select the **Layout template** you want to use (one of the **Landscape** templates is suggested) .



4. Click **O.K.**
5. Maximize the **Layout** screen.

EDIT TITLE

1. **Double Click** on the title ***View1***.
2. **Edit** the title to reflect the purpose of the map, e.g., forestry, and population.
3. Click **O.K.**
4. **Resize** the title using the corner handles and move it into a position that is visually balanced.
5. **Click and Drag** to reposition the legend if necessary.

EDIT NORTH ARROW

1. **Double Click** on the *north arrow* and select a different presentation. Ensure that that the arrow is pointing in the representative north direction of the map.
2. **Click and Drag** to reposition or rotate (using rotation function) as needed.

EDIT SCALE BAR

1. **Double Click** the *scale bar.*

2. **Choose** a new style of scale bar. The scale bar has default properties so you will not be able to change everything to the way you want.
3. **Change** the units to ***kms.***
4. Find a combination of style, scale and divisions that is visually pleasing
5. **Click and Drag** to position to an appropriate location.

ADD LABELS, TEXT, OTHER INFORMATION

1. **Click** on the **Text** tool.
2. **Click** at the position on your layout where you want the text to appear.
3. In the **Text Properties** box, enter your text, e.g., "produced by (your name, date)".
4. **Select** desired formatting information such as **alignment, spacing and rotation.**
5. **Click O.K.** when it appears, as you want it.
6. **Click** on **Pointer** tool.
7. **Select** the new text.
8. **Click once** to change location or size, **Click twice** to change content.

! Experiment with all of the components of the layout until you have the look you want.

CHANGE THE NAME OF THE LAYOUT

You only have to do this if you want to save your project on the computer or on a disk.

1. **Click** menu **Layout.**
2. **Select Properties.**
3. **Change** the **"name".** Click **O.K.**

ARRANGING YOUR LAYOUT

! Experiment with the layout buttons (use the tools page provided at the beginning of this lesson)

- to **group** (tie pieces together so they move as one)
- to **ungroup** (break apart the ties so that individual pieces move)
- **bring to front** (change order that pieces were laid down, e.g., lines on top of polygons)
- **send to back** (change order that pieces were laid down, e.g., polygons behind lines)
- **zoom** (make image bigger or smaller)
- **neat line** (border around layout)

PRINT MAP LAYOUT

1. Click **Print** button.
2. Click **Setup.**
3. **Select** your printer properties as outlined by your teacher.
4. Click **O.K.** to print.

! Note that the map can also be saved to a printer file.